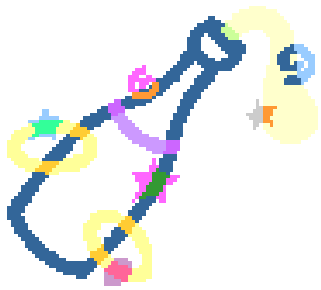
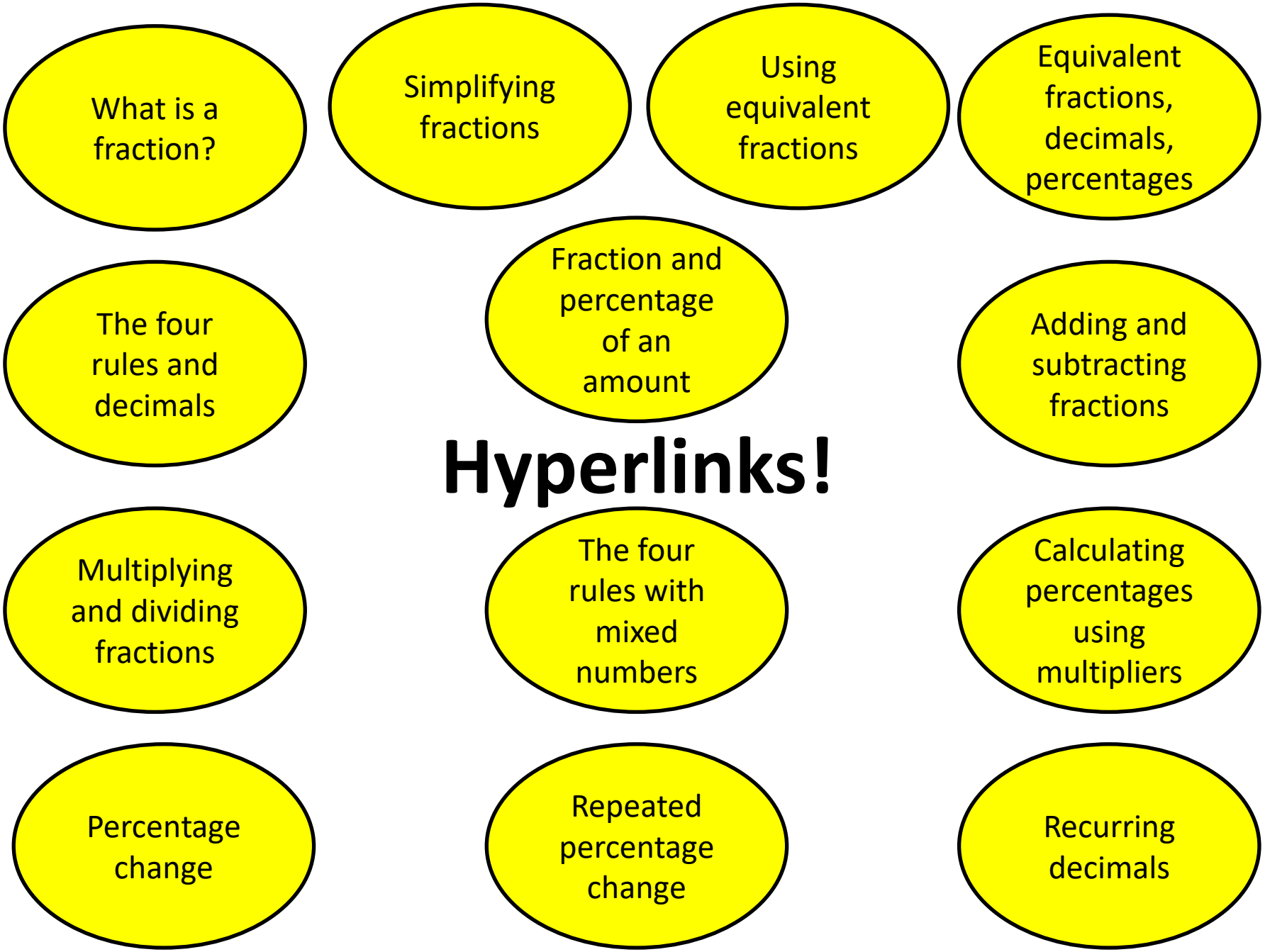
Fractions, Decimals and Percentages



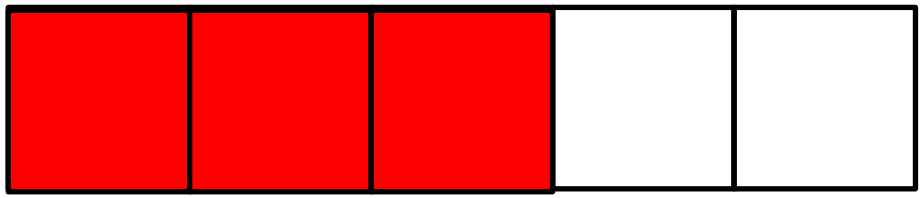

What is a fraction?
Simplifying fractions
Using equivalent fractions
Equivalent fractions, decimals, percentages
The four rules and decimals
Fraction and percentage of an amount
Adding and subtracting fractions
Hyperlinks!
Multiplying and dividing fractions
The four rules with mixed numbers
Calculating percentages using multipliers
Percentage change
Repeated percentage change
Recurring decimals

What is a fraction?

$$\frac{What\ you\ want}{Total}$$

For example:

Shade $\frac{3}{5}$ of this shape:



What you want (shaded) = 3

Total = 5

Simplifying fractions



How to simplify a fraction:

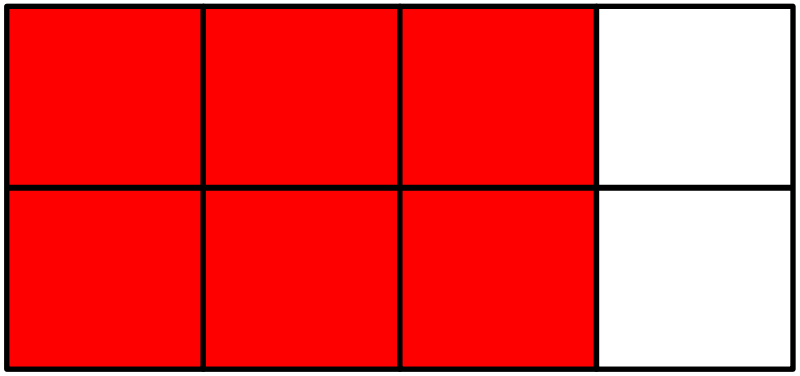
Find a times table that both top (numerator) and bottom (denominator) are in.

Divide both top and bottom by that number.

Keep going until the only times table that goes into both is the 1 times table.

Simplifying using pictures

$\frac{6}{8}$ of this shape is shaded. Simplify the fraction.



I could make the two rows into one.

$$\frac{6}{8} = \frac{3}{4}$$

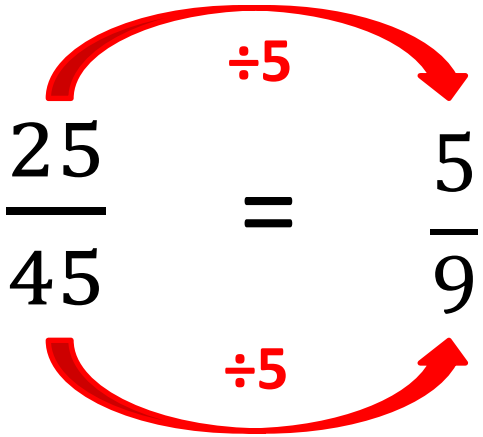
Without a picture, an example:

Simplify $\frac{25}{45}$ to its lowest terms.

What times table are 25 and 45 in?

The 5 times table!

So divide them both by 5.

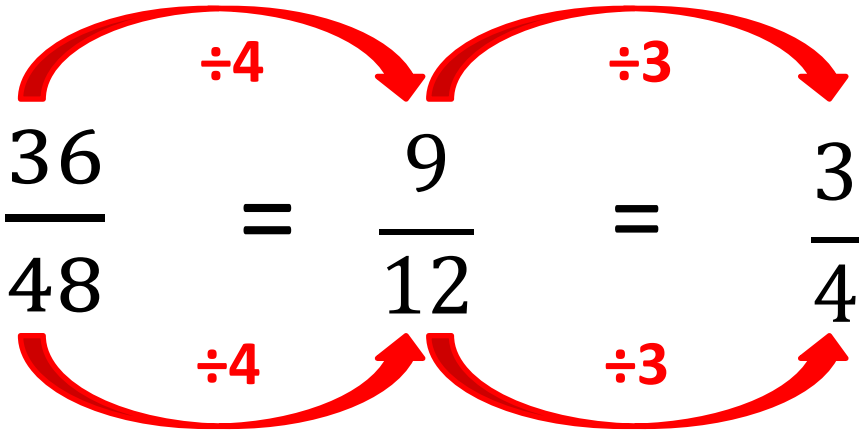


Another example:

Simplify $\frac{36}{48}$ to its lowest terms.

What times table are 36 and 48 in?

They are in lots of tables, so try one and keep going until you can't do any more!



Simplify these, giving your answers in their simplest form:

1. $\frac{24}{30}$ $\frac{4}{5}$
2. $\frac{35}{49}$ $\frac{5}{7}$
3. $\frac{28}{42}$ $\frac{2}{3}$
4. $\frac{16}{48}$ $\frac{1}{3}$
5. $\frac{18}{42}$ $\frac{3}{7}$



Using equivalent fractions



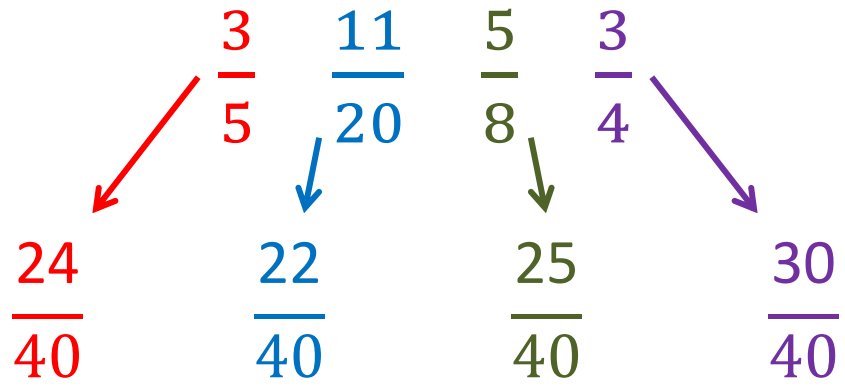
When would you use equivalent fractions?

The main use for them at this stage is to put fractions in order.

If all the fractions you are ordering are over different denominators, it's difficult.

If they are all over the same denominator, it's easy because you just compare the numerators.

Put these fractions in ascending order:



What number appears in all the times tables of the denominators?

Answer: 40, so put them all "over" 40.

Answer: $\frac{11}{20}$ $\frac{3}{5}$ $\frac{5}{8}$ $\frac{3}{4}$

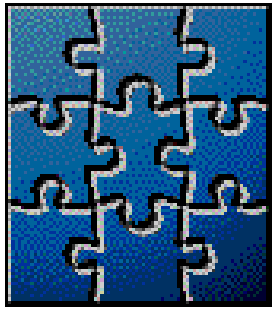
Put these fractions in descending order:

$$\frac{1}{4} \quad \frac{2}{5} \quad \frac{3}{10} \quad \frac{1}{2} \quad \frac{7}{20}$$

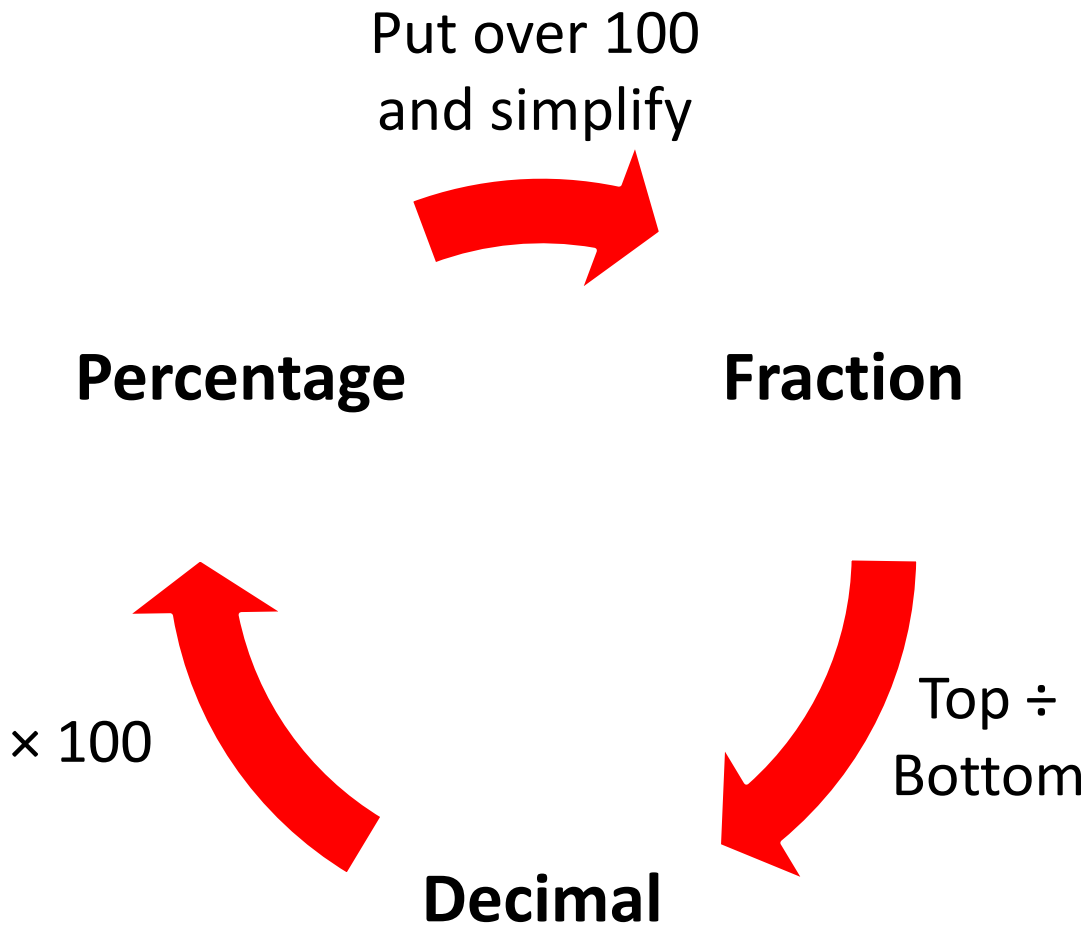
Answer: $\frac{1}{2} \quad \frac{2}{5} \quad \frac{7}{20} \quad \frac{3}{10} \quad \frac{1}{4}$



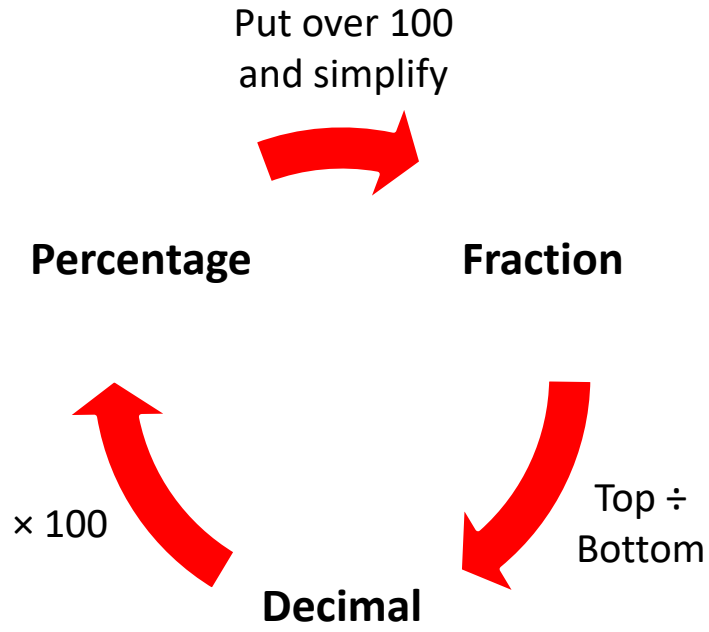
Equivalent fractions, decimals and percentages



The cycle!


Put over 100
and simplify
Fraction
Top ÷
Bottom
Decimal
× 100
Percentage

How does the cycle work?



Write $\frac{3}{8}$ as a percentage and decimal.

As a decimal: 3 ÷ 8 = 0.375

As a percentage: 0.375 × 100 = 37.5%

Back to a fraction: $\frac{37.5}{100} = \frac{375}{1000} = \frac{75}{200} = \frac{3}{8}$

How it's used in exams:

This is a typical exam style question:

Put these in ascending order:

Ascending mean "from smallest to biggest



Convert them all to the same thing – decimals.

Answer: 0.2 $\frac{1}{4}$ 40% 0.5 $\frac{3}{5}$

Have a go at this question:

Put these is order, smallest first:

$\frac{3}{4}$ 0.72 67% $\frac{4}{5}$ 0.82



Answer: 67% 0.72 $\frac{3}{4}$ $\frac{4}{5}$ 0.82

The four rules and decimals



Adding and Subtracting Decimals

The key to this is to make sure that the decimal points are all in a column before you start.

So 1.45 + 12.2 would look like this:

$$\begin{array}{rrrrr} & 1 & . & 4 & 5 \\ 1 & 2 & . & 2 & + \\ \hline 1 & 3 & . & 6 & 5 \end{array}$$

Decimal points in a column

Have a go at these:

1. 16.4 + 3.25 **= 19.65**
2. 22.74 + 1.89 **= 24.83**
3. 14.76 – 2.23 **= 12.53**
4. 8.23 – 3.57 **= 4.66**



Multiplying “easy” decimals:

Count how many decimal places are in the question.

Forget the decimal points and multiply the numbers as if they were whole.

The answer needs the same number of decimal places as in the question.

An example:

Calculate 0.6×0.4



There are 2 decimal places in the question.

Multiply: $6 \times 4 = 24$

There should be 2 decimal places in the answer:

0.24



Have a go at these:

1. 0.3×0.4 **= 0.12**
2. 0.5×0.8 **= 0.4**
3. 0.2×0.3 **= 0.06**
4. 1.2×0.7 **= 0.84**
5. 0.06×0.3 **= 0.018**
6. 0.04×0.006 **= 0.00024**

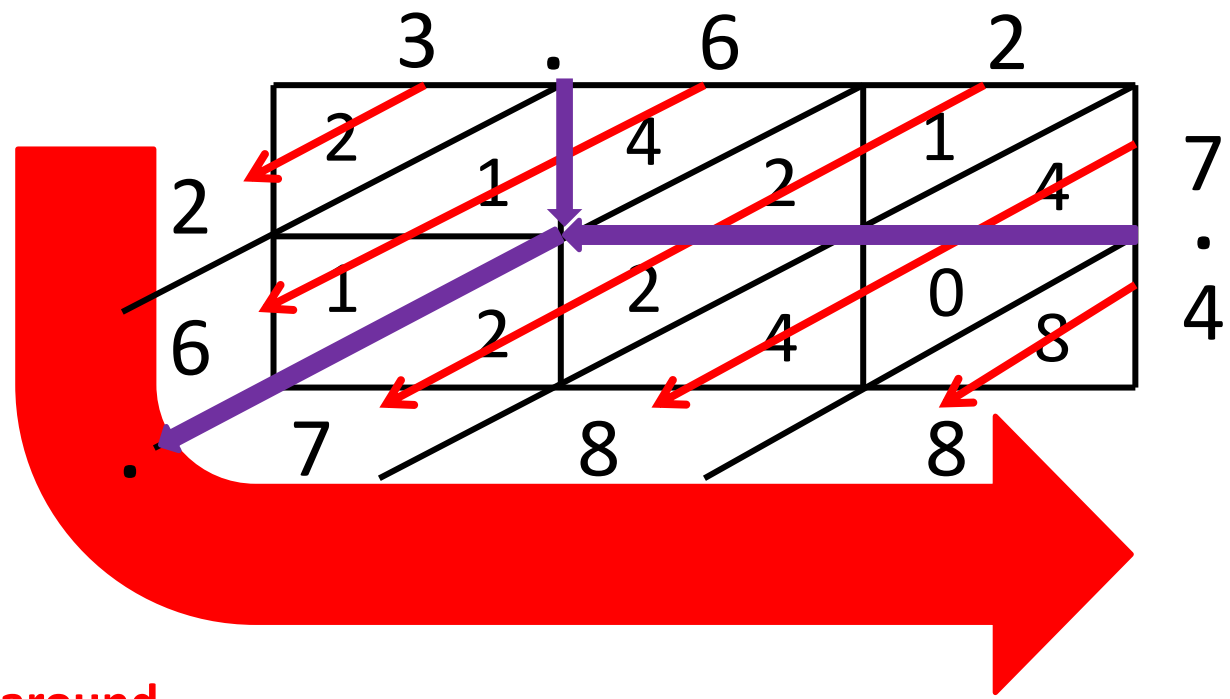


Multiplying decimals

You use a grid, forgetting about the decimal points before replacing them at the end.

An example of multiplying decimals:

Calculate 3.62 × 7.4.



Work out where the decimal point should be

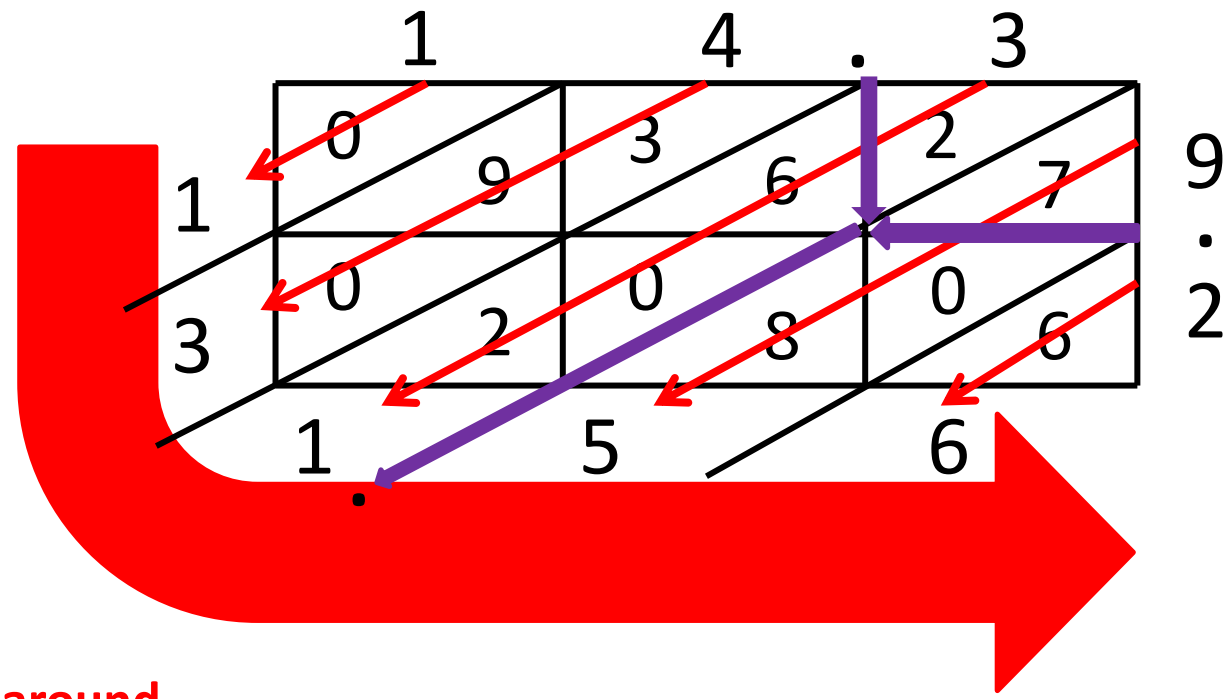
Add up along the diagonals

Read around the bottom

Answer: 26.788

Another example:

Calculate 14.3 × 9.2.



Work out where the decimal point should be

Add up along the diagonals

Read around the bottom

Answer: 131.56

Try these:

1. 34.7×2.6 **= 90.22**
2. 9.42×1.7 **= 16.014**
3. 341×0.46 **= 156.86**



Dividing Decimals

Dividing by a decimal is difficult!

But we can keep multiplying by 10 until the divisor (the number you are dividing by) becomes a whole number.

This makes it much easier!

How do we do that?

If I show you a simple example it should become clear.

$20 \div 5 = 4$

That's easy!

But…

$200 \div 50 = 4$

Multiply the question by 10

Multiply the question by 100

$2000 \div 500 = 4$

Etc.

We keep getting the answer 4

Dividing by a decimal:

Calculate 24 ÷ 0.06

Multiply by 10:

240 ÷ 0.6

Multiply by 10 again:

2400 ÷ 6

The divisor is now a whole number!

2400 ÷ 6 = 400

Have a go at these:

1. $18 \div 0.3$ **= 60**
2. $3 \div 0.06$ **= 50**
3. $0.14 \div 0.7$ **= 0.2**
4. $320 \div 0.008$ **= 40,000**



Fraction and percentage of an amount



Fraction of an amount

Remember this:

Divide by the bottom

Multiply by the top

If you do each calculation that way around you will be working with smaller numbers, but it does work either way.

Some fraction of a number examples:

Example 1

Find $\frac{3}{7}$ of £28.

£28 ÷ 7 = £4

£4 × 3 = £12

Answer: £12

Example 2

Find $\frac{5}{6}$ of £45.

£45 ÷ 6 = £7.50

£7.50 × 5 = £37.50

Answer: £37.50

Have a go at these:

Find:

1. $\frac{1}{6}$ of £30. **= £5**
2. $\frac{3}{4}$ of £24. **= £5**
3. $\frac{2}{5}$ of £45. **= £18**
4. $\frac{5}{7}$ of £140. **= £100**
5. $\frac{3}{8}$ of £36. **= £13.50**



Percentage of an amount

A percentage is a fraction with a denominator of 100.

Divide by the 100

Multiply by the percentage you want

Some percentage of a number examples:

Example 1

Find 30% of £150.

£150 ÷ 100 = £1.50

£1.50 × 30 = £45

Answer: £45

Example 2

Find 15% of £23.

£23 ÷ 100 = £0.23

£0.23 × 15 = £3.45

Answer: £3.45

Could you do this without a calculator?

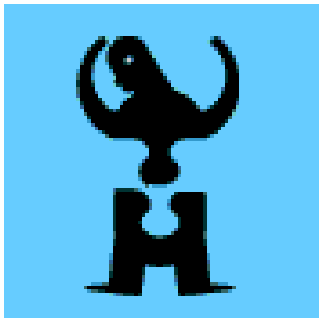
Have a go at these:

Find:

1. 40% of £30. **= £12**
2. 25% of £24. **= £6**
3. 80% of £45. **= £36**
4. 62% of £140. **= £86.80**
5. 7% of £36. **= £2.52**



Adding and subtracting fractions



Important!

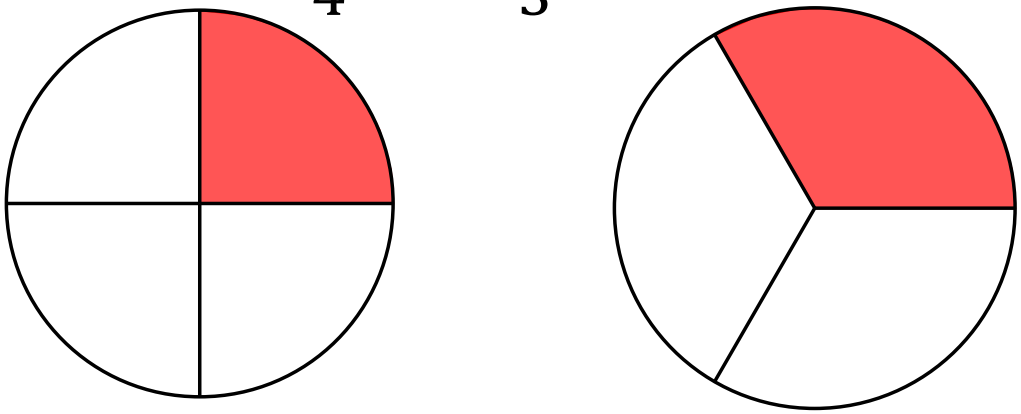
You may have heard the phrase “common denominator” when people talk about adding and subtracting fractions.

What that means is that you can’t add or subtract unless the bottom numbers of all the fractions are the same.

Common Denominators – Eh?

If we look at it in diagrams, it might make more sense.

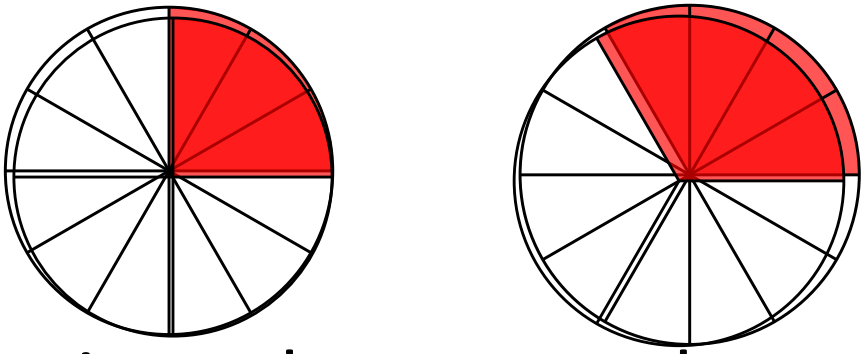
If I wanted to add $\frac{1}{4}$ and $\frac{1}{3}$, it would look like this:



The parts are different sizes.

How common denominators work:

Let's look at $\frac{1}{4}$ and $\frac{1}{3}$ again:



The parts in each are now the same size.

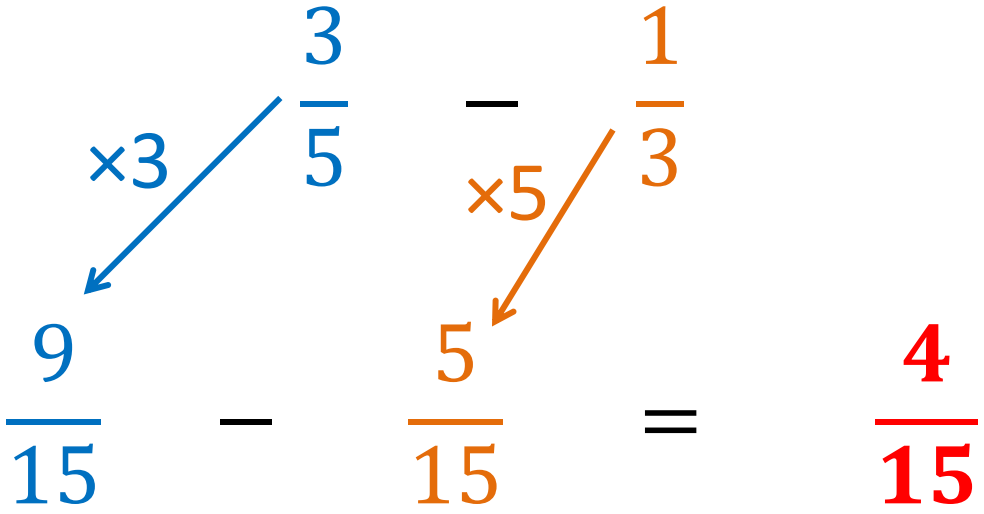
Each shape is now divided into **12 parts**, of which a total of **7** are shaded.

So $\frac{1}{4} + \frac{1}{3} = \frac{\mathbf{7}}{\mathbf{12}}$

Another example, without diagrams:

Calculate $\frac{3}{5} - \frac{1}{3}$.

Common denominator: $5 \times 3 = 15$



Yet another example, without diagrams:

Calculate $\frac{2}{3} + \frac{4}{7}$.

Common denominator: $3 \times 7 = 21$

$$\frac{2}{3} + \frac{4}{7}$$

$\times 7$ $\qquad$ $\times 3$

$$\frac{14}{21} + \frac{12}{21} = \frac{\mathbf{26}}{\mathbf{21}} = \mathbf{1}\frac{\mathbf{5}}{\mathbf{21}}$$

Have a go at these:

1. $\frac{4}{5} - \frac{3}{4}$ $= \frac{1}{20}$

2. $\frac{3}{8} + \frac{1}{4}$ $= \frac{5}{8}$

3. $\frac{9}{10} - \frac{1}{6}$ $= \frac{22}{30} = \frac{11}{15}$

4. $\frac{5}{8} + \frac{2}{5}$ $= 1\frac{1}{40}$

5. $\frac{4}{5} - \frac{3}{7}$ $= \frac{13}{35}$



Multiplying and dividing fractions



Multiplying fractions

This is really easy!

Multiply the top.

Multiply the bottom.

That’s really it!

Two examples

Example 1

Calculate $\frac{3}{5} \times 7$.

When you have a whole number, "put it over 1".

$$\frac{3}{5} \times \frac{7}{1}$$

$$\frac{\mathbf{3 \times 7}}{\mathbf{5 \times 1}} = \frac{\mathbf{21}}{\mathbf{5}} = \mathbf{4\frac{1}{5}}$$

Example 2

Calculate $\frac{4}{9} \times \frac{3}{8}$.

$$\frac{4}{9} \times \frac{3}{8}$$

Multiply top, multiply bottom.

$$\frac{\mathbf{4 \times 3}}{\mathbf{9 \times 8}} = \frac{\mathbf{12}}{\mathbf{72}} = \frac{\mathbf{1}}{\mathbf{6}}$$

Could you have simplified the second example before multiplying?

Try these:

1. $\frac{4}{7} \times 9$ $= \frac{36}{7} = 5\frac{1}{7}$

2. $5 \times \frac{2}{11}$ $= \frac{10}{11}$

3. $\frac{4}{5} \times \frac{3}{7}$ $= \frac{12}{35}$

4. $\frac{8}{15} \times \frac{3}{4}$ $= \frac{2}{5}$

5. $\frac{7}{19} \times \frac{38}{49}$ $= \frac{2}{7}$



Dividing fractions

This is as easy as multiplying!

First you turn the right hand fraction over.

Then multiply.

Two examples

Example 1

Calculate $\frac{3}{5} \div 7$.

Put the whole number over 1.

$$\frac{3}{5} \div \frac{7}{1}$$

Turn the right hand fraction over and multiply:

$$\frac{3}{5} \times \frac{1}{7}$$

$$\frac{3}{5} \times \frac{1}{7} = \mathbf{\frac{3}{35}}$$

Example 2

Calculate $\frac{4}{9} \div \frac{3}{8}$.

Turn the right hand fraction over and multiply:

$$\frac{4}{9} \times \frac{8}{3}$$

$$\mathbf{\frac{32}{27} = 1\frac{5}{27}}$$

Have a go at these:

1. $\frac{4}{7} \div 9$ $= \mathbf{\frac{4}{63}}$

2. $5 \div \frac{2}{11}$ $= \mathbf{\frac{55}{2} = 27\frac{1}{2}}$

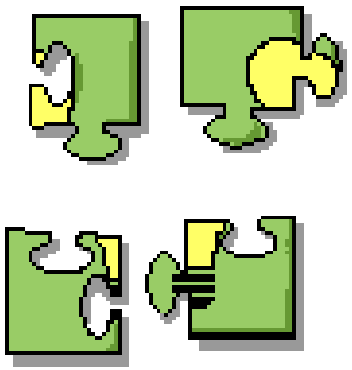
3. $\frac{4}{5} \div \frac{3}{7}$ $= \mathbf{\frac{28}{15} = 1\frac{13}{15}}$

4. $\frac{8}{15} \div \frac{3}{4}$ $= \mathbf{\frac{32}{45}}$

5. $\frac{7}{8} \div \frac{21}{32}$ $= \mathbf{\frac{4}{3} = 1\frac{1}{3}}$



The four rules with mixed numbers



What are mixed numbers?

They are whole numbers and fractions.

Eg. $2\frac{1}{3}$ is a mixed number.

It is easier to convert mixed numbers to improper or top-heavy fractions – you must convert in order to multiply and divide.

Eg. $2\frac{1}{3} = \frac{2\times3+1}{3} = \frac{7}{3}$

Then you add, subtract, multiply and divide as you have done before.

Adding and subtracting

Example 1

Calculate $2\frac{3}{4} + 1\frac{1}{3}$.

Convert to improper:

$$\frac{11}{4} + \frac{4}{3}$$

Common denominator:

$$\frac{33}{12} + \frac{16}{12}$$

Add up:

$$\mathbf{\frac{49}{12} = 4\frac{1}{12}}$$

Example 2

Calculate $2\frac{3}{5} - 1\frac{6}{7}$.

Convert to improper:

$$\frac{13}{5} - \frac{13}{7}$$

Common denominator:

$$\frac{91}{35} - \frac{65}{35}$$

Subtract:

$$\mathbf{\frac{26}{35}}$$

Have a go at these:

1. $2\frac{1}{4} + 1\frac{1}{2}$ $= 3\frac{3}{4}$
2. $2\frac{3}{4} - 1\frac{2}{5}$ $= 1\frac{7}{20}$
3. $5\frac{2}{3} + 1\frac{3}{5}$ $= 7\frac{4}{15}$
4. $3\frac{1}{4} - 1\frac{3}{7}$ $= 1\frac{23}{28}$
5. $2\frac{6}{11} + 3\frac{4}{7}$ $= 6\frac{9}{77}$



Multiplying and dividing

Example 1

Calculate $2\frac{3}{4} \times 1\frac{1}{3}$.

Convert to improper:

$$\frac{11}{4} \times \frac{4}{3}$$

Multiply:

$$\frac{44}{12} = 3\frac{8}{12} = 3\frac{2}{3}$$

Example 2

Calculate $2\frac{3}{4} \div 1\frac{1}{3}$.

Convert to improper:

$$\frac{11}{4} \div \frac{4}{3}$$

Turn the right hand fraction:

$$\frac{11}{4} \times \frac{3}{4}$$

Multiply:

$$\frac{33}{16} = 2\frac{1}{16}$$

Could you have simplified before multiplying?

Have a go at these:

1. $2\frac{1}{4} \times 1\frac{1}{2}$ **$= 3\frac{3}{8}$**

2. $2\frac{3}{4} \div 1\frac{2}{5}$ **$= 1\frac{27}{28}$**

3. $5\frac{2}{3} \times 1\frac{3}{5}$ **$= 9\frac{1}{15}$**

4. $3\frac{1}{4} \div 1\frac{3}{7}$ **$= 2\frac{11}{40}$**

5. $2\frac{6}{11} \div 3\frac{4}{7}$ **$= \frac{196}{275}$**



Calculating percentage change using a multiplier



How we calculate percentage increase and decrease:

You could find the percentage and add or subtract it to the original amount.

Or you could find the “multiplier”.

An increase of 10% means that you have 110% of the original amount.

A decrease of 15% means that you have 85% of the original amount.

How do you use multipliers?

Example 1

Increase £270 by 20%.

An increase of 20% means that I have 120% of the original amount.

120% as a multiplier is 1.2.

£270 × 1.2 = **£324**

Example 2

Decrease £135 by 7%.

A decrease of 7% means that I have 93% of the original amount.

93% as a multiplier is 0.93.

£135 × 0.93 = **£125.55**

Some questions for you:



1. Increase £230 by 30%. **£299**
2. Decrease £65 by 45%. **£42.25**
3. Increase £32 by 4%. **£33.28**
4. Decrease £768 by 17%. **£637.44**
5. Andy has been offered a pay rise. He is currently paid £7.80 per hour. He must choose between a rise of 20p per hour or an increase of 3%. Which should he choose?

An increase of 3% would be £8.03 per hour.

Working backwards – finding the original amount

Whereas before we multiplied…

When finding an original amount, you always divide!

Original amount × multiplier = New amount

Original amount = New amount ÷ multiplier

Finding the original:

Example 1

The population of a town has increased by 15% to 207,000 in the last 10 years. How many people lived in the town 10 years ago?

207,000 ÷ 1.15

180,000 people

Example 2

A shirt was reduced in a sale by 20%. It was £71.20 in the sale. How much is the shirt normally?

£71.20 ÷ 0.8

£89

Find the original prices for these sale items:



1. A pair of trousers reduced by 30% to £77. **£110**
2. A pair of shoes reduced by 25% to £63.75. **£85**
3. A jacket reduced by 42% to £84.10. **£145**

Find the original amounts following increases:



1. A wage increased by 4% to £540.80 per week. **£520**
2. A population increased by 11% to 59,940. **54,000**
3. A picture increased by 71% to 307.8cm. **180cm**

Calculating percentage change



How do we do that?

Remember this:

$$\frac{Difference}{Original\ amount} \times 100$$

This will work for all percentage change questions.

Two examples

Example 1

The population of a town went up from 50,000 to 63,000 in 10 years. What percentage increase is this?

Change = £13,000

Original amount = £50,000

$$\frac{13000}{50000} \times 100 = \mathbf{26}\%$$

Example 2

The value of a car went down from £12,500 to £9,000 in 1 year. What percentage decrease is this?

Change = £3,500

Original amount = £12,500

$$\frac{3500}{12500} \times 100 = \mathbf{28}\%$$

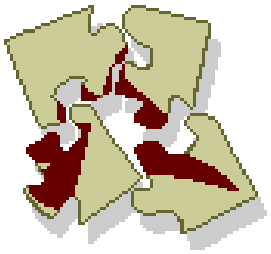
Have a go at this:

A property developer bought a house for £162,000. They spent £7000 renovating the house and then sold it for £185,000.

By what percentage had they increased the value of the property?

Answer: 9.88% (to 2 decimal places)

Repeated percentage change



Repeated percentage change:

I invest £500 at a rate 4% per annum (means "per year"). How much would be in my account after 3 years?

$500 \times 1.04 = £520$

$520 \times 1.04 = £540.80$

$540.80 \times 1.04 = \mathbf{£562.43}$

There is a quicker way!

The calculation on the left is this:

$£500 \times 1.04 \times 1.04 \times 1.04$

We can write that like this:

$£500 \times 1.04^3 = \mathbf{£562.43}$

A couple of examples:

Example 1

I invest £850 at a rate 2% per annum. How much would be in my account after 6 years?

$£850 \times 1.02^6$

Answer: £957.24

Example 2

The value of a car reduces by 15% per year. If the car cost £18,000 when new, what would if be worth after 5 years?

$£18{,}000 \times 0.85^5$

Answer: £7986.70

Two questions for you:

Question 1

The profit of a company has been projected to increase by 7% per year. This year's profit was £230,000. How much profit do they expect to get in 4 years time?

Answer: £301,483.08



Question 2

Dave is reducing his credit card bill by 15% per month having built up a debt of £12,000. How many months will it take him to get his debt under £5000?

Answer: 6 months

Writing fractions as recurring decimals and vice versa



What is a recurring decimal?

A recurring decimal never ends and repeats a number or sequence of numbers.

We use the recurring notation like this:

$$0.\dot{6} = 0.6666666666\ldots$$

$$0.\dot{7}\dot{2} = 0.7272727272\ldots$$

$$0.3\dot{4} = 0.3444444444444\ldots$$

$$0.\dot{5}872\dot{5} = 0.587258725872\ldots$$

Fractions to recurring decimals:

As with all fractions, to convert a fraction to a decimal you just divide the top (numerator) by the bottom (denominator).

So, to convert $\frac{4}{9}$ to a decimal I would do this:

$$
\begin{array}{r}
0.4444 \\
\hline
9 \;\big|\; 4.^{4}0^{4}0^{4}0^{4}0
\end{array}
$$

It will continue forever, so write it as $\mathbf{0.\dot{4}}$

Convert these to decimals without a calculator using dot notation:

1. $\frac{2}{3}$ $\quad 0.\dot{6}$
2. $\frac{1}{6}$ $\quad 0.1\dot{6}$
3. $\frac{4}{11}$ $\quad 0.\dot{3}\dot{6}$
4. $\frac{2}{7}$ $\quad 0.\dot{2}85714\dot{2}$



Recurring decimals to fractions:

The key is to get rid of the recurring part.

We do this by multiplying by powers of 10 then subtracting.

The power of 10 we multiply by depends on how many numbers recur.

Converting from decimals to fractions

Example 1

Write $0.\dot{2}$ as a fraction.

One number is recurring

$$n = 0.\dot{2}$$
$$10n = 2.\dot{2}$$

The recurring parts are the same so subtract:

$$9n = 2$$

$$\boldsymbol{n = \frac{2}{9}}$$

Example 2

Write $0.\dot{3}\dot{7}$ as a fraction.

Two numbers are recurring

$$n = 0.\dot{3}\dot{7}$$
$$100n = 37.\dot{3}\dot{7}$$

The recurring parts are the same so subtract:

$$99n = 37$$

$$\boldsymbol{n = \frac{37}{99}}$$

Try converting these to fractions:

1. $0.\dot{7}$ $\mathbf{\frac{7}{9}}$

2. $0.\dot{4}\dot{3}$ $\mathbf{\frac{43}{99}}$

3. $0.\dot{6}4\dot{1}$ $\mathbf{\frac{641}{999}}$



A little trickier, but the same process:

Example 1

Write $0.4\dot{7}$ as a fraction.

Don't need this one anymore

$$n = 0.4\dot{7}$$
$$10n = 4.\dot{7}$$
$$100n = 47.\dot{7}$$

The recurring parts are the same so subtract:

$$90n = 43$$

$$\boldsymbol{n = \frac{43}{90}}$$

Example 2

Write $0.5\dot{2}4\dot{3}$ as a fraction.

Don't need this one anymore

$$n = 0.5\dot{2}4\dot{3}$$
$$10n = 5.\dot{2}4\dot{3}$$
$$10000n = 5243.\dot{2}4\dot{3}$$

The recurring parts are the same so subtract:

$$9990n = 5238$$

$$\boldsymbol{n = \frac{5238}{9990}}$$

Have a go at converting these to fractions:

1. $0.6\dot{7}$ $\quad \mathbf{\frac{61}{90}}$

2. $0.5\dot{7}\dot{1}$ $\quad \mathbf{\frac{566}{990} = \frac{283}{495}}$

3. $0.59\dot{3}\dot{1}$ $\quad \mathbf{\frac{5872}{9900} = \frac{1468}{2475}}$



SUCCESS CRITERIA – WHERE ARE WE NOW?

Level	Learning outcomes:	R	A	G
F1	I can simplify fractions to their lowest terms.			
E2	I can order fractions by using equivalence.			
E2	I can use fraction, decimal and percentage equivalents.			
E3	I can use the four rules with decimals.			
E3	I can find a fraction and percentage of a number.			
D2	I can add and subtract fractions.			
C3	I can multiply and divide fractions.			
C1	I can add, subtract, multiply and divide mixed numbers.			
C3	I can increase or decrease a value by a given percentage using a multiplier.			
C3	I can find the percentage change.			
C1	I can perform repeated percentage changes using just one calculation.			
B2	I can write fractions as recurring decimals and vice versa.			